

bessel expansion method (from paper)

$$j_1 := 2.404825576958 \quad j_2 := 5.5200781102863 \quad j_3 := 8.653727912911 \quad j_4 := 11.791534439014$$

$$j_5 := 14.930917708488 \quad j_6 := 18.071063967911 \quad j_7 := 21.211636629879$$

$$n := 7 \quad s2 := 1..n \quad s := 0..n \quad c := 40 \quad zt := 40 \quad zb := -40 \quad V(z, r) := 5 \quad g(z) := \frac{\pi \cdot (2 \cdot z - zb - zt)}{zt - zb}$$

$$a_{1,s} := \frac{2 \cdot \int_{zb}^{zt} V(z, c) \cdot \cos\left[\frac{1}{2} \cdot (2 \cdot s + 1) \cdot g(z)\right] dz}{(zt - zb) \cdot 10 \left[\frac{(2 \cdot s + 1) \cdot \pi \cdot c}{2 \cdot (zt - zb)} \right]} \quad a_{2,s} := \frac{2 \cdot \int_{zb}^{zt} V(z, c) \cdot \sin((2 \cdot s) \cdot g(z)) dz}{(zt - zb) \cdot 10 \left[\frac{4 \cdot s \cdot \pi \cdot c}{(zt - zb)} \right]}$$

$$a_{3,s2} := \frac{2 \cdot \int_0^c V(zb, r) \cdot J_0\left(\frac{j_{s2} \cdot r}{c}\right) \cdot r dr}{c^2 \cdot \sinh\left[\frac{(zt - zb) \cdot j_{s2}}{c}\right] \cdot J_1(j_{s2})^2} \quad a_{4,s2} := \frac{2 \cdot \int_0^c V(zt, r) \cdot J_0\left(\frac{j_{s2} \cdot r}{c}\right) \cdot r dr}{c^2 \cdot \sinh\left[\frac{(zt - zb) \cdot j_{s2}}{c}\right] \cdot J_1(j_{s2})^2}$$

$J_0(j_{s2})$
$-1.4087 \cdot 10^{-14}$
$-4.19883 \cdot 10^{-15}$
$3.28839 \cdot 10^{-15}$
$-6.55076 \cdot 10^{-14}$
$-4.39971 \cdot 10^{-14}$
$1.4693 \cdot 10^{-14}$
$4.51704 \cdot 10^{-14}$

$$h(s) := \frac{(2 \cdot s + 1) \cdot \pi \cdot (zt + zb)}{zt - zb}$$

$$m(s) := \frac{2 \cdot s \cdot \pi \cdot (zt + zb)}{zt - zb}$$

$$C_0 := \sum_s \left[a_{1,s} \cdot \cos(h(s)) - a_{2,s} \cdot \sin\left[\frac{2 \cdot s \cdot \pi \cdot (zt + zb)}{zt - zb}\right] \right] \dots \quad a^T = \begin{bmatrix} 0 & 5.48687 & 0 & 0 & 0 \\ 0 & -0.71921 & 0 & 0.13058 & 0.13058 \\ 0 & 0.11998 & 0 & -1.7084 \cdot 10^{-4} & -1.7084 \cdot 10^{-4} \\ 0 & -0.02134 & 0 & 2.5918 \cdot 10^{-7} & 2.5918 \cdot 10^{-7} \\ 0 & 0.00394 & 0 & -4.17942 \cdot 10^{-10} & -4.17942 \cdot 10^{-10} \\ 0 & -7.43159 \cdot 10^{-4} & 0 & 6.9678 \cdot 10^{-13} & 6.9678 \cdot 10^{-13} \\ 0 & 1.42466 \cdot 10^{-4} & 0 & -1.18628 \cdot 10^{-15} & -1.18628 \cdot 10^{-15} \\ 0 & -2.76207 \cdot 10^{-5} & 0 & 0 & 0 \end{bmatrix}$$

$$C_1 := \frac{\pi \cdot c}{zt - zb} \cdot \left[\sum_s \left[a_{1,s} \cdot (2 \cdot s + 1) \cdot \sin(h(s)) + 4 \cdot a_{2,s} \cdot s \cdot \cos(m(s)) \right] \right] \dots$$

$$+ \sum_{s2} \left(-a_{3,s2} \cdot \cosh\left(\frac{zt \cdot j_{s2}}{c}\right) + a_{4,s2} \cdot \cosh\left(\frac{zb \cdot j_{s2}}{c}\right) \right) \cdot j_{s2}$$

$$C_2 := \left(\frac{\pi \cdot c}{zt - zb} \right)^2 \cdot \left[\sum_s \left[-\frac{1}{2} \cdot a_{1,s} \cdot (2 \cdot s + 1)^2 \cdot \cos(h(s)) + (-8) \cdot a_{2,s} \cdot s^2 \cdot \sin(m(s)) \right] \right] \dots$$

$$+ \left(-\frac{1}{2} \right) \cdot \sum_{s2} \left(a_{3,s2} \cdot \sinh\left(\frac{zt \cdot j_{s2}}{c}\right) - a_{4,s2} \cdot \sinh\left(\frac{zb \cdot j_{s2}}{c}\right) \right) \cdot (j_{s2})^2$$

$$C_3 := \left(\frac{\pi \cdot c}{zt - zb} \right)^3 \cdot \left[\sum_s \left[\frac{1}{6} \cdot a_{1,s} \cdot (2 \cdot s + 1)^3 \cdot \sin(h(s)) + \left(\frac{32}{3} \right) \cdot a_{2,s} \cdot s^3 \cdot \cos(m(s)) \right] \right] \dots$$

$$+ \left(\frac{1}{6} \right) \cdot \sum_{s2} \left(a_{3,s2} \cdot \cosh\left(\frac{zt \cdot j_{s2}}{c}\right) - a_{4,s2} \cdot \cosh\left(\frac{zb \cdot j_{s2}}{c}\right) \right) \cdot (j_{s2})^3$$

$$C_4 := \left(\frac{\pi \cdot c}{zt - zb} \right)^4 \cdot \left[\sum_s \left[\frac{1}{24} \cdot a_{1,s} \cdot (2 \cdot s + 1)^4 \cdot \cos(h(s)) + \left(-\frac{332}{3} \right) \cdot a_{2,s} \cdot s^4 \cdot \sin(m(s)) \right] \right] \dots$$

$$+ \left(\frac{1}{24} \right) \cdot \sum_{s2} \left(a_{3,s2} \cdot \sinh\left(\frac{zt \cdot j_{s2}}{c}\right) - a_{4,s2} \cdot \sinh\left(\frac{zb \cdot j_{s2}}{c}\right) \right) \cdot (j_{s2})^4$$

$$C_5 := \left(\frac{\pi \cdot c}{zt - zb} \right)^5 \cdot \left[\sum_s \left[\frac{1}{120} \cdot a_{1,s} \cdot (2 \cdot s + 1)^5 \cdot \sin(h(s)) + \left(-\frac{128}{15} \right) \cdot a_{2,s} \cdot s^5 \cdot \cos(m(s)) \right] \right] \dots$$

$$\begin{aligned}
& \left(\frac{zt-zb}{120} \right) \cdot \left[\sum_s \left[\frac{-1}{120} \cdot \left(a_{3,s2} \cdot \cosh \left(\frac{zt \cdot j_{s2}}{c} \right) - a_{4,s2} \cdot \cosh \left(\frac{zb \cdot j_{s2}}{c} \right) \right) \cdot (j_{s2})^5 \right. \right. \\
C_6 := & \left(\frac{\pi \cdot c}{zt-zb} \right)^6 \cdot \left[\sum_s \left[\frac{-1}{720} \cdot a_{1,s} \cdot (2 \cdot s + 1)^6 \cdot \cos(h(s)) + \left(-\frac{256}{45} \right) \cdot a_{2,s} \cdot s^6 \cdot \sin(m(s)) \right] \right] \dots \\
& + \left(\frac{-1}{720} \right) \cdot \sum_{s2} \left(a_{3,s2} \cdot \sinh \left(\frac{zt \cdot j_{s2}}{c} \right) - a_{4,s2} \cdot \sinh \left(\frac{zb \cdot j_{s2}}{c} \right) \right) \cdot (j_{s2})^6
\end{aligned}$$

RESULTS

$$\begin{array}{llll}
C_0 = 6.262974380171849 & C_2 = -5.048682231844674 & C_4 = -2.23787 & C_6 = 0.73229 \\
C_1 = 0 & C_3 = 0 & C_5 = 0 &
\end{array}$$

TEST

$$\begin{aligned}
\text{phi}(z,r) := & C_0 \cdot 1 + \frac{C_1}{c} \cdot (z) + \frac{C_2}{c^2} \cdot \left(-\frac{1}{2} \cdot r^2 + z^2 \right) + \frac{C_3}{c^3} \cdot \left(-\frac{3}{2} \cdot r^2 \cdot z + z^3 \right) + \frac{C_4}{c^4} \cdot \left(\frac{3}{8} \cdot r^4 - 3 \cdot r^2 \cdot z^2 + z^4 \right) \dots \\
& + \left[\frac{C_5}{c^5} \cdot \left(\frac{15}{8} \cdot r^4 \cdot z - 5 \cdot r^2 \cdot z^3 + z^5 \right) \right] + \frac{C_6}{c^6} \cdot \left(-\frac{5}{16} \cdot r^6 + \frac{45}{8} \cdot r^4 \cdot z^2 - \frac{15}{2} \cdot r^2 \cdot z^4 + z^6 \right)
\end{aligned}$$

$$\begin{array}{ll}
\text{phi}(0,0) = 6.26297 & V(0,0) = 5 \\
\text{phi}(1,1) = 6.2614 & V(1,1) = 5
\end{array}$$